

your computers, the next step is to get the right cable. When networking just two computers, you can connect them directly using an RJ-45 Crossover cable, which you can buy from any computer dealer. If you have three or more PCs, you should consider investing in a switch that will allow you to easily connect all the PCs using regular RJ-45 cables. A wireless network will need your computers to be Wi-Fi-enabled, which is not a problem if you have Wi-Fi capable laptops, but if you have desktop PCs, you will need to buy wireless networking PCI cards. There are two ways in which you can connect computers wirelessly: in Ad-Hoc mode or by using an access point to assign fixed IP addresses. More on this later.

11.3 Setting Up The Network

Let's first talk about IP addresses. What is an IP address? IP addresses are nothing but names of computers represented in numeric terms. In a TCP/IP network the names are represented in 32-bit numeric address written in four parts and each part is divided by dots. Each number in this part can be in the range of 0 to 255, for example, 192.192.0.1

In a home network you can assign IP address randomly as long as each IP address is unique. If two IP addresses on a network are identical, they will clash and neither will work correctly. For example, if a system has the IP 192.192.0.1, another system cannot use the same IP and will have to be set as 192.192.0.2 or 192.192.0.3. The Internet Assigned Numbers Authority has reserved the following three blocks for private networks such as your LAN: 10.0.0.0 to 10.255.255.255, 172.16.0.0 to 172.31.255.255 and 192.168.0.0 to 192.168.255.255. So if you are setting up a LAN that's also connected to the Internet, you can safely use these numbers within your network without causing errors in DNS (Domain Name System) resolving.

When you type a Web page address such as www.thinkdigit.com into your browser, the browser contacts a Domain Name System (DNS) server and resolves the IP address of the computer it refers to.